DISTRIBUTED AND STREAMING BASE CLIENT-SERVER COLLABORATION APPLICATION

Course: Network Application Development

Duration: 6 Weeks (Take-home Project)

Project Type: Individual / Pair Project

I. INTRODUCTION

1. CURRENT PRACTICES

With the rapid global shift toward remote work, distributed teams, and decentralized systems, reliance on robust, real-time network applications has never been higher. Modern organizations require seamless data exchange over local and wide-area networks, while navigating challenges like latency, packet loss, and data security. Understanding how to build resilient network software from the ground up is a critical skill for today's software and network engineers.

2. THE NEED FOR THIS APPLICATION

While commercial solutions (like Slack, Microsoft Teams, or cloud storage) exist, they often route data through centralized third-party servers, which can raise privacy, compliance, and bandwidth concerns for closed-network environments. There is a distinct need for a lightweight, secure, and self-hosted network application that allows users on the same network (or VPN) to communicate and share data directly without relying on external corporate infrastructure.

3. GOAL OF THE APPLICATION

The goal of this project is to build a **Distributed and Streaming-based Client-Server Collaboration Application**. Students will design and implement a custom application layer protocol over UDP STREAMING that enables real-time messaging, active user discovery, and reliable file transfer between multiple clients over a network. Due to the native requirement of messaging application, **reliable communication** must be provided.

4. Features of the Application

- **User Authentication & Discovery:** Users can log in, broadcast their presence, and see a live list of other active peers on the network.
- **Streaming-based Messaging:** Support for both broadcast (global chat) and unicast (private direct messaging). Refer **Steam GameNetworkingSocket** and **Google QUIC** for message-oriented transport layer.
- **Reliable File Transfer:** The ability to send and receive files of various formats and sizes without data corruption.

- **Connection Resilience:** The app gracefully handles sudden client disconnects, timeouts, and network interruptions.
- **Secure Data Transmission:** Basic encryption applied to the payloads to prevent packet-sniffing vulnerabilities.

5. Target Users

- Internal corporate teams operating on secure, air-gapped, or isolated local area networks (LAN).
- Students or study groups needing a fast, peer-to-peer way to collaborate and share files without internet access.
- Privacy-conscious individuals who prefer self-hosted communication tools.

6. Techniques and Technologies Integrated

- **Socket Programming:** Implementation of TCP (for reliable file/message transfer) and UDP (for heartbeat/discovery broadcasts).
- **Concurrency:** Multi-threading or Asynchronous I/O to handle multiple simultaneous client connections without blocking the main thread.
- **Application-Layer Protocol Design:** Designing custom packet structures (Headers, Payloads, End-of-Message flags).
- **Sequencing and ordering**: Maintaining monotonic sequence number with window-based sliding window for flow control and in-order delivery.
- **Acknowledgement and Retransmission:** Efficient retransmitting of only missing segments.
- **Congestion control:** Support streaming specific with BBR / CUBIC streaming support.
- **Cryptography:** Integration of TLS/SSL sockets or symmetric encryption (e.g., AES) for data payloads.
- **Serialization:** Encoding and decoding structured data (JSON, XML, or custom binary packing) over the wire.

7. Design Requirements

- **Architecture:** Must clearly separate network logic, data processing, and the user interface.
- **Time-sensitive streaming**: The system must provide an async mechanism to fetch data from a stream without any. The streaming only handles chunk of bytes in UDP and leaves your protocol logics in application layer in which you eliminate Head-of-line (HoL) blocking.
- **User Interface:** Can be a clean Command Line Interface (CLI) or a Graphical User Interface (GUI). It must be intuitive and display network status clearly.
- **Scalability:** The server/tracker must be capable of handling at least 10-20 concurrent clients efficiently.

8. Technical Requirements

- **Programming Language:** Python, Java, C++, or Go (Instructor to confirm restricted languages, if any).

- **No High-Level Networking Libraries:** Students must use raw sockets (socket library in Python, java.net.Socket in Java, etc.). HTTP frameworks (like Flask/Django) or WebSockets are strictly prohibited to ensure core networking concepts are practiced.
- **Environment:** Must run successfully on standard operating systems (Windows, macOS, or Linux) using localhost or local IP addressing.

9. Development Process & Grading Rubric (10 Points Total)

This project is broken down into a 10-step development lifecycle. Each successfully implemented step is worth **1 point**.

Step	Milestone	Score
1	**Setup & Protocol Design:** Define the custom packet structure, message types (e.g., AUTH, MSG, FILE), and state diagrams in documentation before coding.	1 pt
2	**Basic Client-Server Handshake:** Implement basic TCP socket connection where a client can connect to the server and receive an acknowledgment.	1 pt
3	**Authentication & State Management:** Implement a login mechanism. The server must track connected users, their IP addresses, and active ports.	1 pt
4	**Concurrency Implementation:** Integrate multi-threading or **Async** I/O so the server can handle multiple clients connecting and disconnecting simultaneously.	1 pt
5	**Time-sensitive streaming:** Implement the routing logic for clients to send messages to the server, and the server to broadcast or forward them to the correct recipient.	1 pt
6	**Reliable File Transfer:** Implement file serialization, chunking (for large files), and reassembly over UDP without data corruption.	1 pt
7	**Network Resilience & Error Handling:** Implement try/except blocks to handle socket timeouts, broken pipes, and graceful client exits without crashing the server.	1 pt
8	**Security Layer:** Encrypt the data payloads (using a standard cryptographic library) before sending, and decrypt upon receipt.	1 pt

Step	Milestone	Score
9	**HTTP based UI with client-side JavaScript only:** Build a user-friendly interface that clearly displays incoming messages, file transfer progress bars, and active user lists.	1 pt
10	**Testing & Code Quality:** Code must be cleanly formatted, commented, and pass local network simulation tests (connecting from multiple terminal windows).	1 pt
	TOTAL	**10 pts**

10. Submission Guidelines

Required Submission Content:

Students must submit a .zip file or a link to a private GitHub repository containing the following:

1. **Source Code:**
    - All client and server scripts.
    - A README.md file with strict, step-by-step instructions on how to install dependencies, run the server, and connect the clients.
2. **Technical Explanation Report (PDF):**
    - **System Architecture:** A high-level diagram showing how clients and the server interact.
    - **Protocol Specification:** A breakdown of the custom packet/message structures designed for the application.
    - **Sequence Diagram:** A flow chart showing the lifecycle of a message and a file transfer.
    - **Challenges & Solutions:** A brief 1-page reflection on network challenges faced (e.g., resolving deadlocks, handling packet fragmentation) and how they were solved.